Valuation Scheme

25900529

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMOS201 - USER INTERFACE AND USER EXPERIENCE FOR APP DEVELOPMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Multiple Choice Answers

1. c	**6. d**
2. b	**7. a**
3. b	**8. a**
4. b	**9. a**
5. b	**10. b**

Part B
Short Essay Questions (5*4=20)

11. Define responsive design and explain its importance in Android apps.

Definition - 2 marks

Responsive design means designing an application interface so that it automatically adjusts to different screen sizes, resolutions, and orientations such as phones, tablets, and foldable devices.

Importance - 2 marks

Importance in Android apps:

- Android devices come in many screen sizes
- Ensures consistent user experience everywhere
- Improves readability and usability

- Avoids broken layouts and scrolling issues
- Makes apps future-proof for new devices

Example -1 mark

Example: An Android app that adjusts its layout when switching from portrait to landscape and fits correctly on both mobile phones and tablets is using responsive design. Design.

12. Explain the role of search patterns in improving user experience in mobile apps.

Definition – 2 marks

Search patterns are design structures used in mobile apps to help users find information or features quickly using elements like search bars, filters, voice search, and suggestions.
They guide users in locating content efficiently without navigating multiple screens.

Importance – 2 marks

- Saves user time by quickly showing relevant information
- Enhances navigation, especially in content-heavy apps (e-commerce, social media)
- Reduces user frustration and improves satisfaction
- Increases accessibility and usability
- Helps users discover features or content they didn't know existed

Example – 1 mark

Example: In Amazon app, the top search bar with filters and suggestions helps users quickly find products without browsing categories.

13. Explain the difference between low-fidelity and high-fidelity prototypes.

Definition – 2 marks

- Low-fidelity prototype: A basic, rough visual sketch of an interface showing layout and concept without detailed design or interactivity.
- High-fidelity prototype: A detailed, polished, and interactive version of the interface closely resembling the final app design.

Difference – 2 marks

Low-Fidelity Prototype	**High-Fidelity Prototype**
Rough sketches / wireframes	Polished UI with colors, fonts, icons
Low detail, low interaction	Fully interactive and realistic
Used in early idea stage	Used for advanced testing & client demos
Quick to create	Time-consuming to build

Example – 1 mark

Example: A paper sketch of app screens is low-fi, while an interactive Figma design with real buttons and transitions is high-fi.

14. Why are visual design elements (colors, typography, icons) important in mobile UI?

Definition – 2 marks

Visual design elements like colors, typography, and icons create the look & feel of a mobile UI. They help users understand information, improve usability, and enhance the app experience.

Importance – 2 marks

- Colors guide attention and communicate meaning (e.g., red for error, green for success)

- Typography improves readability and hierarchy of text
- Icons visually represent actions and reduce text usage
- Improves visual consistency and brand identity

Example – 1 mark

Example: WhatsApp uses green theme, simple Rybo font, and familiar icons like chat, call, camera for easy navigation.

15. What are some fundamental UI design rules that ensure a user-friendly app interface?

Definition – 2 marks

UI design rules are guidelines that ensure an app interface is simple, usable, and easy for users to interact with.

Key Rules – 2 marks

- Keep interface clean and simple
- Use consistent colors and icons
- Ensure readable text and proper spacing
- Provide easy navigation and clear buttons
- Maintain visual hierarchy (headings > sub-headings > body text)

Example – 1 mark

Example: Google apps follow Material Design — clean UI, consistent buttons, and readable fonts.

16. List commonly used Android widgets and explain their purpose.

Definition – 2 marks

Android widgets are UI components used to build app interfaces and allow user interaction.

Common Widgets & Purpose – 2 marks

- Button – triggers actions (submit, next)
- TextView – displays text
- EditText – takes input from user
- ImageView – displays images/icons
- RecyclerView – displays list data efficiently

Example – 1 mark

Example: Login screen uses EditText for username/password and Button for Login.

Part C

Essay Questions (10*2=20)

17. Discuss the role of typography, icons, and transitions in improving user experience in Android apps. Provide examples.

Definition – 2 marks

Typography, icons, and transitions are visual elements used to improve readability, simplify navigation, and provide smooth user interaction in mobile apps.

Explanation – 6 marks

- Typography
    - Defines text hierarchy (title, subtitle, body)
    - Improves readability & clarity
    - Creates brand consistency
- Icons

    - Replace text with easy visual cues
    - Save screen space
    - Provide universal meaning (Home, Search, Settings)
- Transitions
    - Smooth animation between screens
    - Helps users understand navigation flow
    - Makes UI feel modern and interactive

Example – 2 marks

Example: Instagram uses bold headings, simple icons, and smooth page transitions for an intuitive experience.

18. Describe navigation patterns in mobile apps and explain their significance in guiding users effectively.

Definition – 2 marks

Navigation patterns help users move between screens and access app features easily.

Explanation – 6 marks

Common navigation types:

- Bottom navigation bar – 3 to 5 main destinations
- Hamburger menu (drawer) – for secondary items
- Tabs – switching between related sections
- Gestures – swipe to move between screens
- Search navigation – find content fast

Significance:

- Helps users locate features quickly
- Reduces confusion & improves usability
- Improves flow & user satisfaction

Example – 2 marks

Example: YouTube uses bottom tabs (Home, Shorts, Subscriptions) + top search bar for easy navigation.

19. Describe the purpose of frames in design tools and how they help organize content on a mobile screen.

Definition – 2 marks

Frames (in Figma/XD) are containers for UI elements like buttons, images, and text to structure app screens.

Explanation – 6 marks

- Hold multiple UI elements together
- Help create screens in fixed sizes (mobile, tablet)
- Support auto-layout & responsive design
- Keep elements aligned & spaced properly
- Used for prototyping scrolling screens

Example – 2 marks

Example: A frame representing a mobile screen (360×800) contains header, menu, buttons, and images arranged neatly.

20. Evaluate the role of transparency and overlays in SVG graphics for modern web interfaces. How can overlay techniques impact performance, accessibility and aesthetic appeal?

Definition – 2 marks

Transparency and overlays in SVG graphics allow UI elements to appear partially see-through. They help layer content visually, create depth, highlight focus areas, and improve interface aesthetics. Overlays are often used for pop-ups, menus, cards, tooltips, and media controls.

Explanation of Role – 4 marks

- Improve focus on content by dimming the background (e.g., modal pop-ups)
- Create depth & hierarchy (frosted glass, blur, shadow overlays)
- Enhance readability when text is placed over images (dark layer behind text)
- Used in modern UI trends like glass-morphism, blurred backgrounds, and hero banners

Impact on Aesthetic Appeal – 1 marks

- Makes UI modern, clean, premium-looking
- Adds smooth transitions and visual depth
- Creates soft, elegant visual separation instead of harsh blocks

Example: Apple's iOS uses blurred glass overlays for control center and menus.

Impact on Performance – 1 mark

- Heavy transparency, blur filters, or multiple SVG layers can increase GPU load
- May affect scrolling and animation smoothness on low-end devices
- Solution: use light opacity layers or PNG fallback for older devices

Impact on Accessibility – 1 mark

- Overlays must maintain contrast so text remains readable
- Should not hide essential content without user control (e.g., ESC to close)
- Screen readers must detect and focus on overlay content
- Poorly designed transparency can confuse visually-impaired users

Example: Netflix uses a dark overlay on thumbnails to keep text readable.

Example – 1 mark

Example: A login pop-up in a shopping app dims the background with a semi-transparent overlay so the form stays in clear focus, improving usability and creating a premium look.